MA321 Quiz 3 Review: Time Series

Name: _____

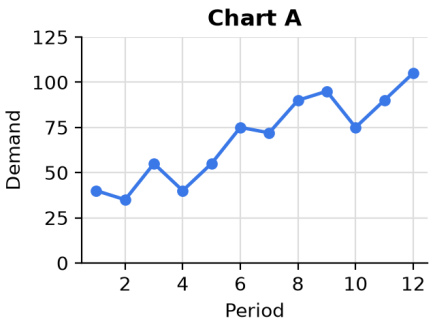
Calculator allowed. No notes, no phone, no internet. Show your work.

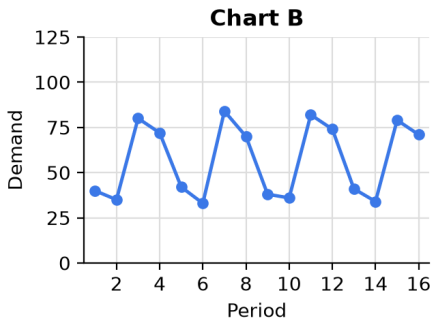
1. Here are the actual demand and the forecast for 3 months. Find the MAPE.

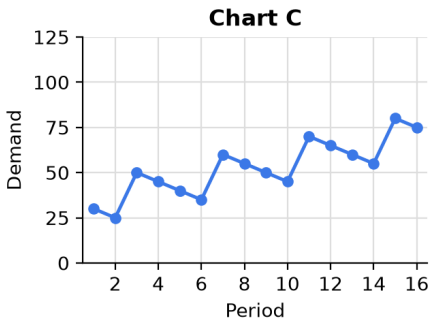
Month	**Actual**	**Forecast**			
1	50	45			
2	80	88			
3	40	50			

MAPE = _____

2. Answer True or False for each.







a) True / False: Chart B is stationary.

b) True / False: Chart A has seasonality based on quarters.

c) True / False: Chart C has a trend.

d) True / False: Chart A has more noise than Chart C.

e) True / False: Chart C is stationary.

3. Suppose you have an average baseline of 20 units per month. The seasonal indices for the next 6 months are 0.6, 1.2, 0.95, 1.1, 1.4, 0.75.

a) Forecast the next 6 months.

b) Forecast the next 6 months if your average baseline were 20% lower.

4.

a) Forecast 1 has a MAPE of 30% and forecast 2 has a MAPE of 12%. Which one is more accurate?

b) Describe in words how the graph of the forecast compares to the graph of the actuals when the MAPE is 3%.

c) True or False: MAPE can be over 100%.

5. Here are the actual sales and the seasonal indices for 6 months.

Month	**Actual**	**Index**	**Deseasonalized**
January	36	0.8	
February	54	0.9	
March	50	1.0	
April	60	1.2	
May	63	1.5	
June	30	0.6	

a) Find the deseasonalized demand for each month and fill it in.

b) Compared to expectations, which month was the best overall and which was the worst? Explain.